

Materials first

Two entirely new doors for the LEFT bay only. Existing pair: 257 x 216 cm; reference, not a verified opening.

PLANNING ISSUE - DO NOT CUT OR ORDER CUSTOM PARTS

Final opening sizes, joint / hinge details and safe jamb attachment must be checked before fabrication. No existing door parts are reused.

ID	Timber, panels and finish - click product names	Qty	Each	Total
A	KD Douglas fir 2x4x8 ft 14 frame / brace boards + 2 for blocking / spare.	16	\$8.00 A	\$128.00
B	15/32 in. BC plywood, 4x8 ft Exterior stamp required; two half-panels per leaf.	4	\$42.37 P	\$169.48
C	1/4 in. BC plywood, 4x8 ft Interior removable lining; verify Exterior classification.	4	\$31.50 P	\$126.00
D	1x4x8 ft pine, 5-pack 20 boards: face trim, lower X, stops, astragal, spare.	4	\$54.83 P	\$219.32
E	1-1/2 in. Comfortboard 80 48 sq ft pack; confirm availability and cut nesting.	1	\$90-120 A	\$90-120
F	Titebond III, 16 oz Candidate only: exterior-door glue approval required.	2	\$9.98 P	\$19.96
G	#8 x1-1/4 in. exterior screws, 1 lb Skins / lap retention; not hinges or wind anchors.	3	\$10.97 P	\$32.91
H	#8 x2 in. exterior screws, 1 lb About 20 mm frame engagement; verify actual layers.	1	\$10.97 P	\$10.97
J	Exterior primer, 1 gal All faces and cut edges; exclude glue mating faces.	1	\$40.98 P	\$40.98
K	Exterior white semi-gloss, 1 gal Two coats; verify label coverage and curing time.	1	\$51.98 P	\$51.98
L	Paintable exterior sealant, 10.1 oz Trim joints / compatible seal interfaces.	2	\$6.98 P	\$13.96

P = publicly displayed Home Depot price checked 12 Sep 2026. A = estimator allowance, not a store quote. Prices are USD, before tax and delivery; no local store selected.

Hardware & project budget

Buy the complete approved hardware system, not just the visually similar items linked below.

ID	Hardware - linked candidates, not approvals	Qty	Each	Total
M	24 in. strap ONLY, N248-047 CANDIDATE. Compatible pintles and bolts separate.	6	\$29.10 P	\$174.60
N	Matching pintles + rated fixings Quote pending; complete hinge assembly / jamb design.	1	\$100-180 A	\$100-180
P	9 ft jamb / header seal Candidate; test outward-swing clearance.	3	\$15.57 P	\$46.71
Q	Meeting-edge gasket, 10 ft Candidate thickness; size to actual compression gap.	1	\$6.97 P	\$6.97
R	Continuous long bottom sweeps Quote pending. At least actual leaf width; 48 in. too short.	2	\$30-60 A	\$60-120
S	18 in. cane bolt Candidate for passive-leaf top / bottom; verify orientation.	2	\$18.93 P	\$37.86
T	Deluxe latch set Candidate; confirm inside release, reach and thickness.	1	\$32.97 P	\$32.97
U	6-1/2 in. pull handle One per face per leaf; use frame-backed fixing.	4	\$8.22 P	\$32.88
V	Positive wind hold-open pair Quote pending. No friction kick-down stops.	1	\$60-120 A	\$60-120
W	Flashing / sleeves / consumables Drip detail, masking, abrasive, brushes, backing seal.	1	\$65-110 A	\$65-110

Materials + hardware allowance: \$1,521-\$1,796

Add 15% materials contingency (\$228-\$269), tools / rentals (\$200-\$450 if needed), and about 7% tax reserve. Planning total: **\$2,085-\$2,691**. Tax reserve is not a tax-rate determination.

Excludes labor, delivery, permit / professional fees, hazardous-material work, jamb or structural repairs, glazing, and right-bay work. Hardware quotes can change this range materially.

Tools & release checks

One small checkpoint before each phase; all unresolved questions are collected on the last page.

Measure / prepare	Cut / assemble	Finish / install
Tape, pencil, 1.8 m level, square, straightedge, winding sticks or straight battens. Moisture meter useful.	Circular saw with guide; supported cutting bench; sharp chisel / mallet or router with depth stop. Drill, bits, countersink, driver.	Random-orbit sander with dust extraction; brushes / roller; caulk gun; sockets / spanners.
Eye and hearing protection; dust protection selected for the material / task.	At least 4 suitable clamps, two long enough for a leaf; flat cauls; rated trestles and solid blocking.	Helper(s), rated lifting / door-support equipment, wedges and a way to secure the opening overnight.

CHECK A / BEFORE DISTURBING THE OLD DOORS

Confirm local requirements with Westfield Building & Construction. Assess unknown old paint and suspect materials before scraping, drilling or sanding. Do not disturb the header, center post or siding.

CHECK B / BEFORE CUTTING THE NEW DOORS

Select the complete hinge / seal system using a conservative weight estimate; its offsets set gaps and blocking. Have an exterior-door carpenter review the measured opening, frame, joints, adhesive and fasteners for this exposure.

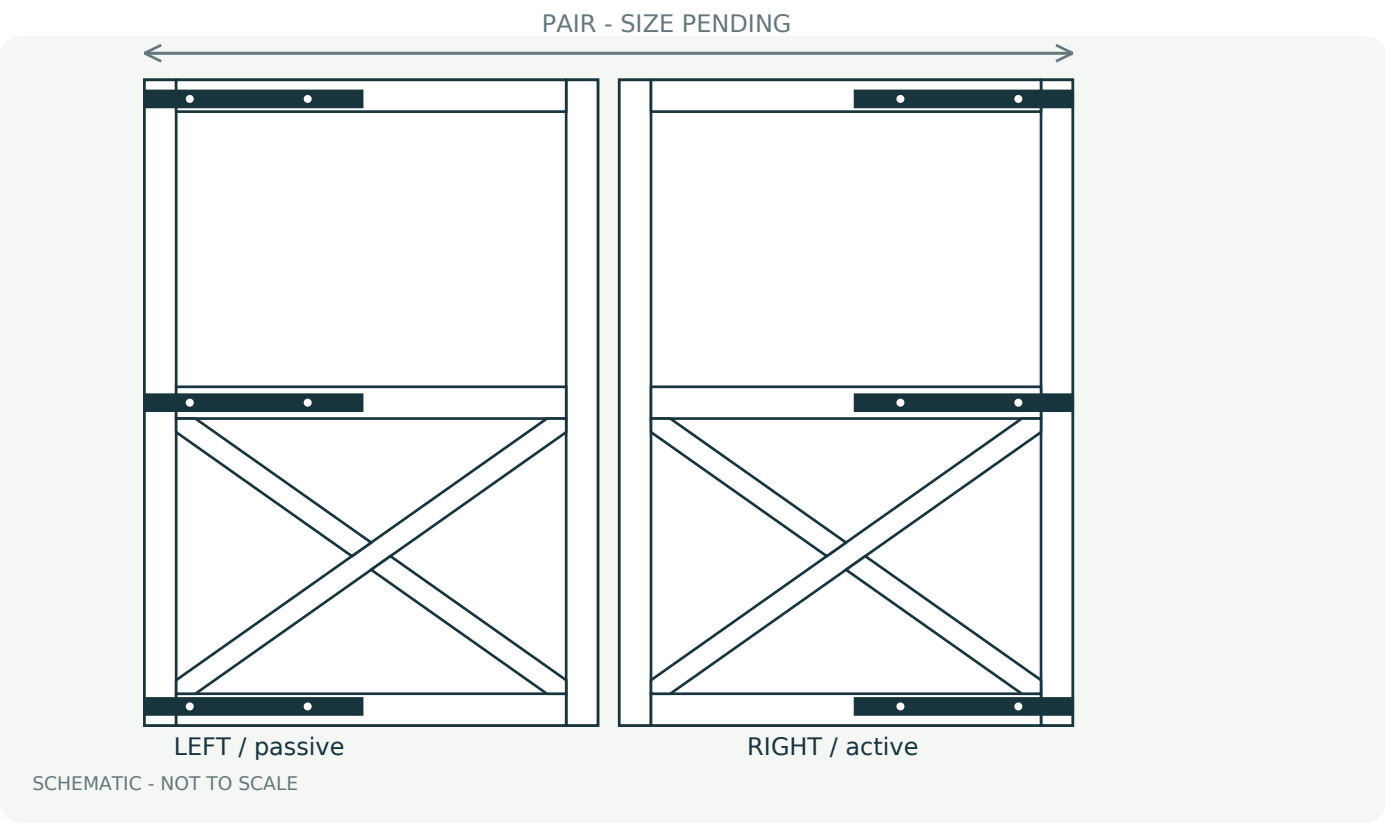
CHECK C / BEFORE HANGING

Have the hinge supplier / installer confirm the full hinge system for leaf width, finished weight, thickness, wind restraint and actual jamb timber. Hardware holes and attachment layout come from that approved detail.

This is a fabrication-planning manual, not a tested, wind-rated, engineered or fire-rated door system. The checks above are genuine release conditions; accurate dimensions alone do not establish structural capacity.

What we are building

Solid upper panels, a lower carriage-style X, concealed internal braces and a shallow insulated cavity.



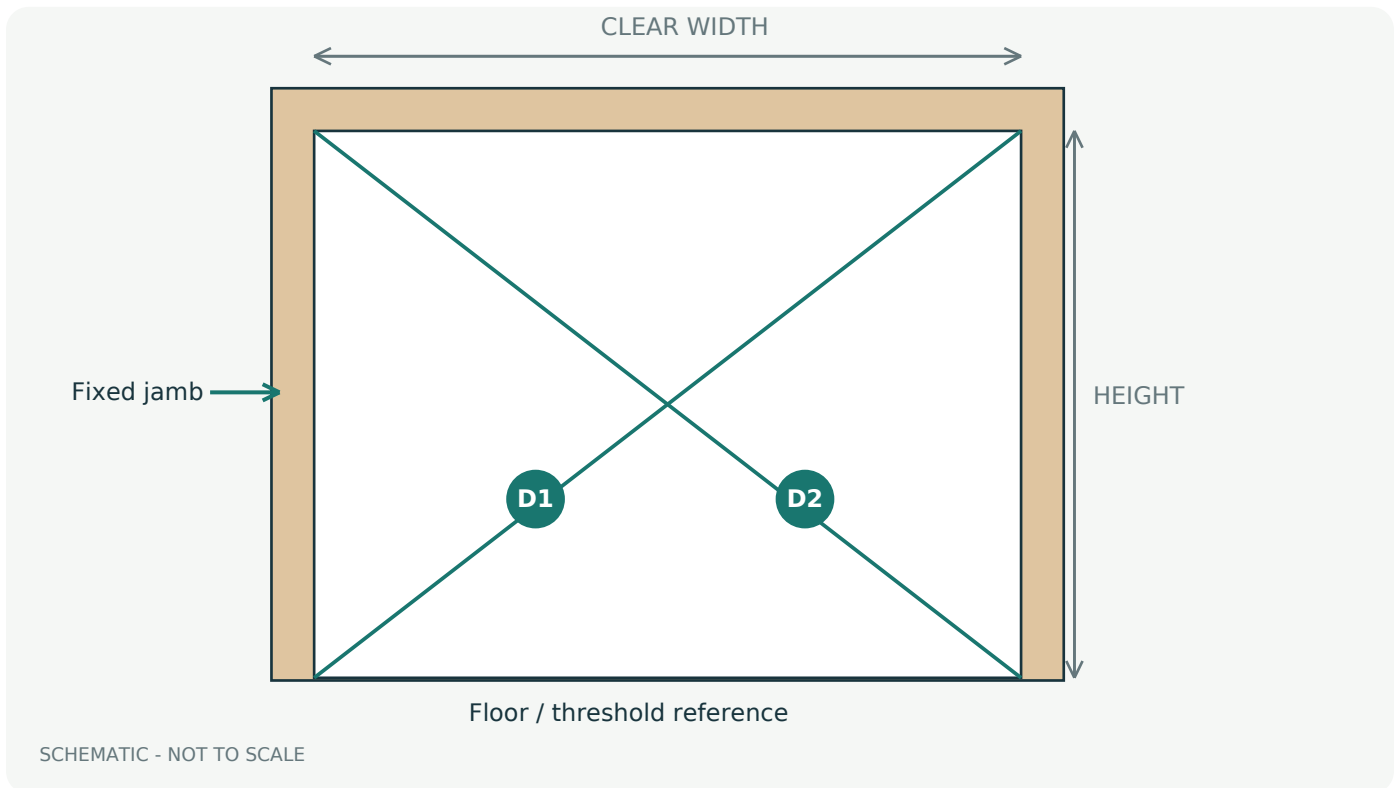
View from driveway: both leaves swing outward. The right leaf opens first; an exterior astragal fixed to it overlaps the left leaf. The original center structural post between garage bays stays untouched.

Outside to inside	Planning thickness / purpose
Painted pine trim	About 19 mm; appearance and seam cover, not the anti-sag brace.
Exterior plywood skin	15/32 in. (about 12 mm), screwed to the frame.
Flat 2x4 frame + stone wool	Actual about 38 x 89 mm timber, lying flat; 38 mm deep cavity.
Removable inner plywood	1/4 in. (about 6 mm); screwed on, not glued.

Approximate thickness: 56 mm without face trim; 75 mm at trim. Verify actual products. Insulation is a nonstandard custom-door use: no whole-door R-value or fire rating is claimed. This version has no windows; adding glazing requires a revised frame, safe glazing detail, weight check and budget.

01 / Survey the opening

Do this before cutting. Keep the old leaves installed while measuring wherever practical.



- 1 Measure clear width at top, middle and bottom; height at left, center and right. Measure between fixed surfaces, not over the old leaves or trim.
- 2 Measure both opening diagonals. Check jambs for plumb, header for level and the mounting faces for twist. A mismatch needs investigation, not a skewed door cut.
- 3 Mark each outward swing on the driveway. Use a level / straightedge to find the highest obstruction across the full swing and decide the final threshold / sweep arrangement.

RECORD, THEN RELEASE

2570 x 2160 mm is the OLD DOOR PAIR, not W and H below. The 140 mm center post and 120 mm side / top timber widths are observations, not proof of depth, soundness or capacity.

Size & cut worksheet

All manufacturing dimensions are in millimeters. Leave blanks until CHECK B is complete.

Symbol	Meaning / formula	Final value
W	Usable corrected clear width at the new door plane.	_____ mm
H	Usable corrected clear height above the chosen floor / threshold reference.	_____ mm
gL / gR / gM	Left / right / meeting gaps from the selected hinge and seal detail.	___ / ___ / ___
gT / gB	Head / bottom gaps; bottom clears highest point throughout the swing.	___ / ___
L	Each equal leaf width = $(W - gL - gR - gM) / 2$.	_____ mm
D	Leaf height = $H - gT - gB$. Use only after floor / header geometry is resolved.	_____ mm
b / t	Actual frame face width / thickness, nominally 89 / 38.	___ / ___ mm
e	Plywood joint gap at the middle rail; provisional 3 mm, confirm panel guidance.	_____ mm

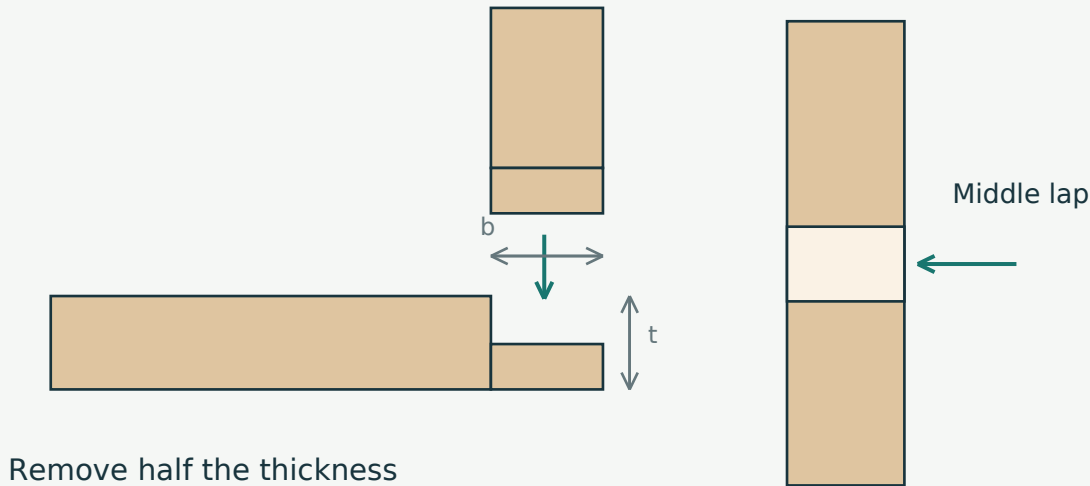
Part / pair quantity	Finished size or method
A1 - stiles / 4	D long; half-laps at top, middle and bottom rail positions.
A2 - rails / 6	L long, full width; end laps b long and t/2 deep. Middle rail centered at D/2.
A3 - braces / 4	Scribe in each assembled half-frame. No fixed angle / final length yet.
B - outer half-panels / 4	L wide x $(D - e)/2$ high; seam centered on the middle rail.
C - inner half-panels / 4	Same envelope as B; field-fit after seal and hardware clearance check.
D - decorative trim	Stiles D; horizontal pieces L - 2 x actual trim width. X pieces scribed.

Check nesting before purchase: rotated half-panels must fit within a 2440 x 1220 mm sheet with saw kerf and edge trimming. This 4-sheet-per-skin allowance applies only while $L < 2440$ and $(D - e)/2 < 1220$. No dimensions on this page are released cut sizes.

02 / Cut the frame & half-laps

Parts A1 + A2. Work on a supported bench with the saw / router manufacturer's safety instructions.

Practice on two offcuts first



SCHEMATIC - NOT TO SCALE

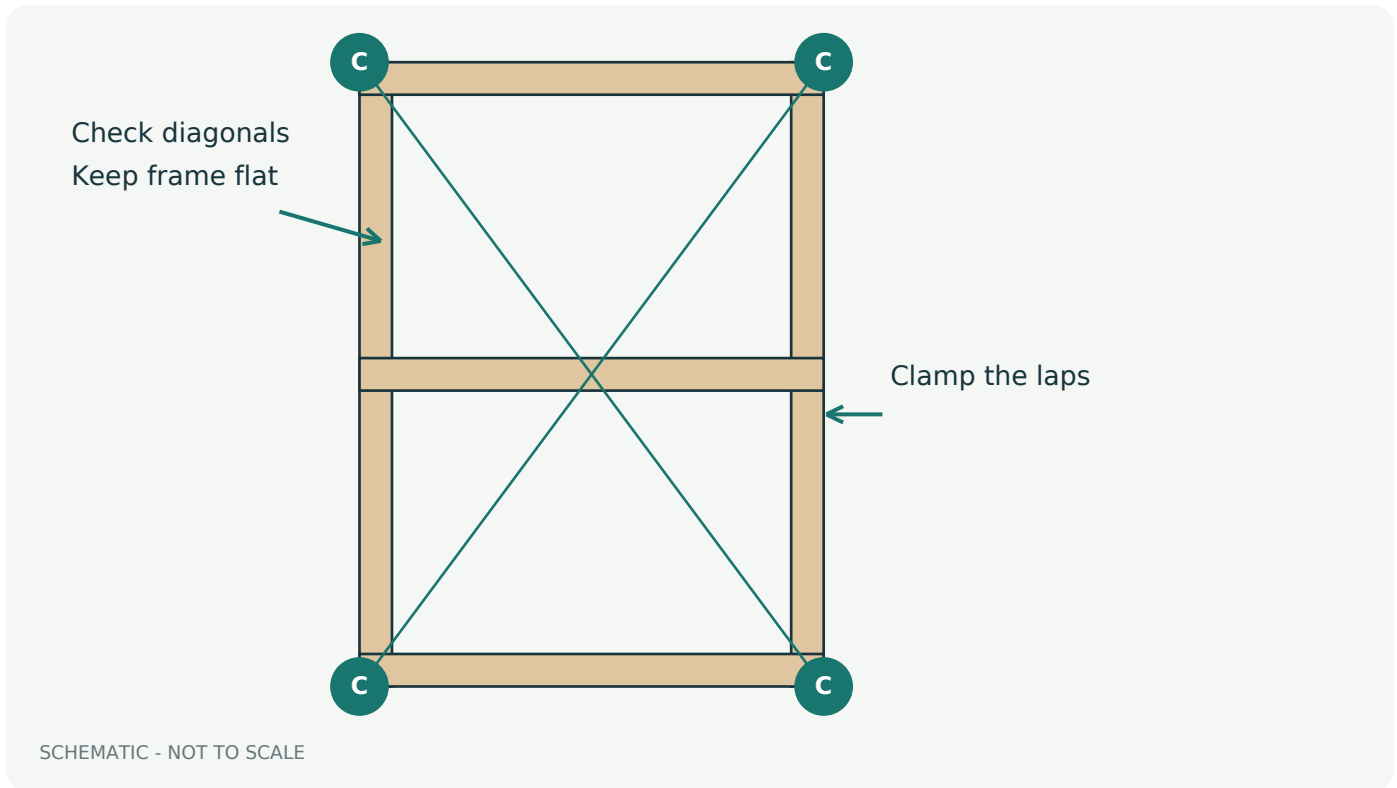
- 1 Select straight, dry, sound boards without cracks or loose knots at the joints or hinge locations. Measure actual b and t . Reject badly bowed or twisted pieces.
- 2 Cut four stiles and six rails to the released worksheet. Label each leaf and keep paired pieces together. Mark all lap faces so rails and stiles interlock flush.
- 3 On scrap, set cutting depth to $t/2$. Make shallow passes, then pare flat. Cut corner laps and the middle-rail housings; test-fit every joint without forcing it.

DONE WHEN

Mating faces contact fully and the assembled frame stays one thickness. Do not use glue or filler to bridge badly cut joints.

03 / Square & assemble each frame

Parts A1 + A2 + F + G. Flat frame first; braces come next.



- 1 Dry-assemble one frame on a verified flat bench. Check both diagonals, outside dimensions and flatness. Adjust until the frame is square without twist.
- 2 After CHECK B accepts the exterior-door adhesive / joint detail, spread the approved glue on clean, unpainted lap faces. Reassemble, clamp with flat cauls, and recheck diagonals before the glue sets.
- 3 For the planning joint detail, predrill and retain each lap with two staggered #8 x 1-1/4 in. screws from the lap side; verify no tip protrudes. Final joint / screw acceptance is part of CHECK B.
- 4 Keep the frame supported and clamped for the adhesive's stressed-joint cure requirement. Allow at least 24 hours before loading, or longer if the product / conditions require.

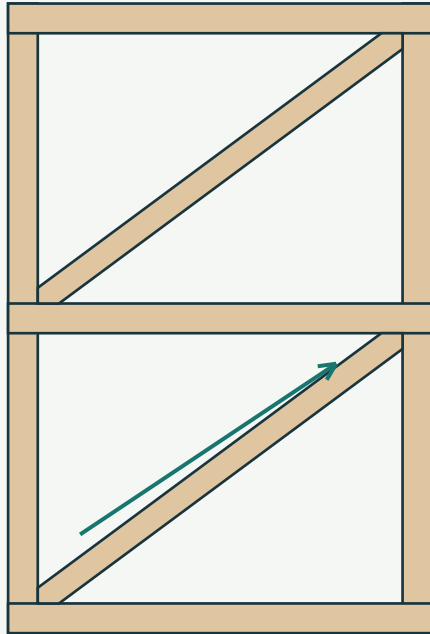
GLUE LIMIT

Titebond III excludes structural / load-bearing use and directs exterior-door users to technical support. Obtain approval or revise the adhesive / joint detail before assembly; the screws shown are not a rated connection.

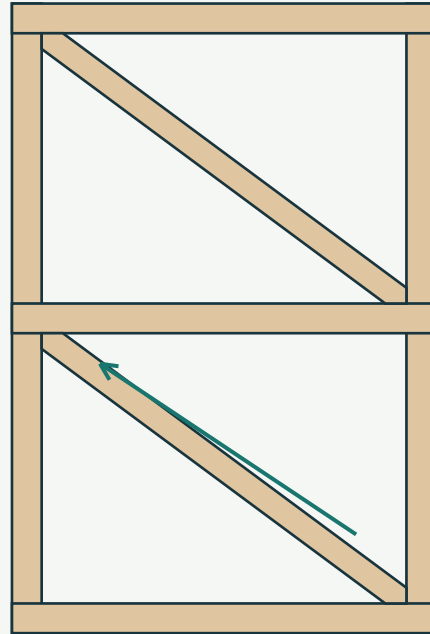
04 / Fit the internal anti-sag braces

Part A3. Outside view, skin omitted. The brace direction follows the hinge side, not the decorative X.

LOWER HINGE CORNER -> UPPER MEETING-SIDE CORNER



Hinges left



Hinges right

SCHEMATIC - NOT TO SCALE

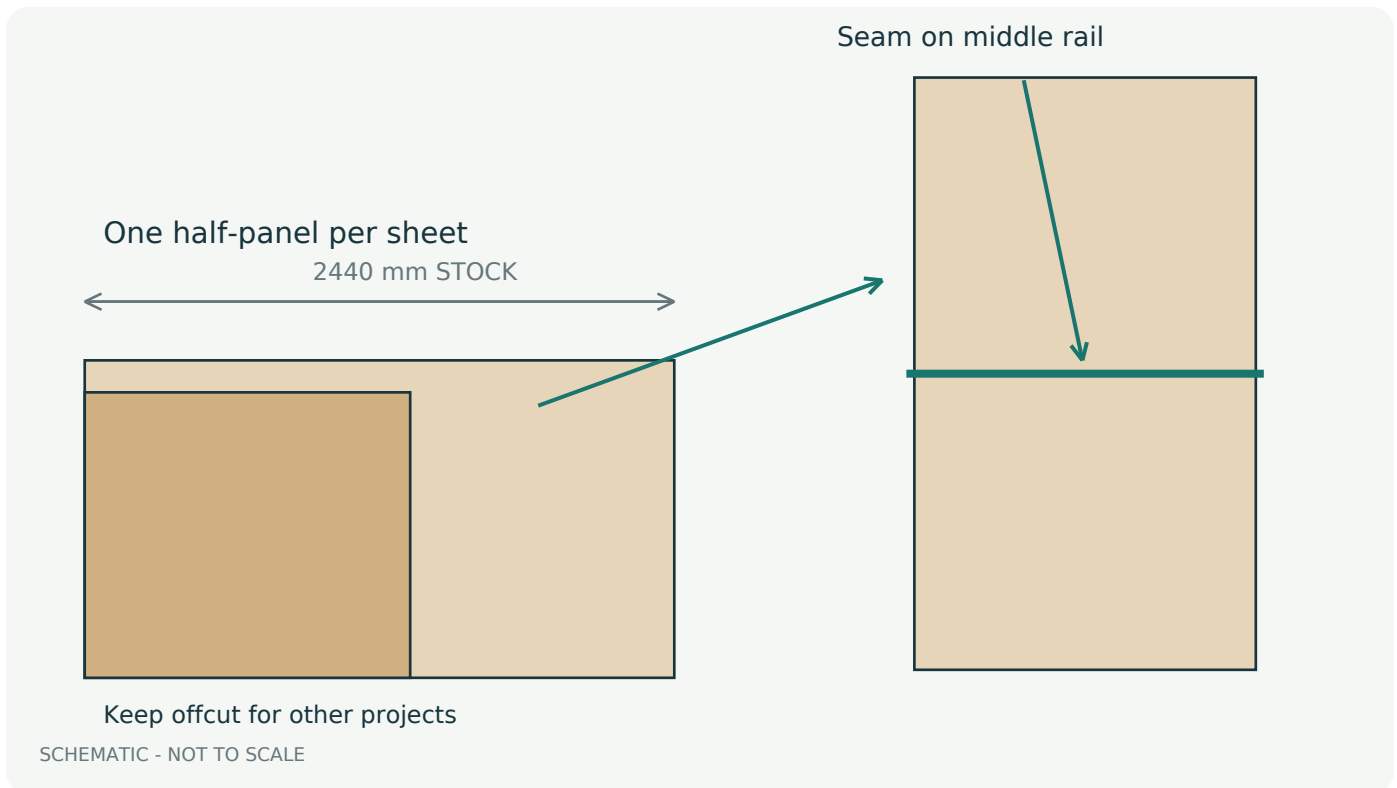
- 1 Fit one brace in each half of the frame: lower hinge-side corner to the opposite upper corner. Mirror the arrangement for the other leaf.
- 2 Lay the brace stock over the opening, mark the actual intersections, then cut and test-fit. Form snug bearing faces against the stile / rail corners; do not copy an assumed 45-degree angle.
- 3 Glue the bearing faces and hold the braces flush with the frame using temporary clamps / cleats. Keep the frame square while curing; the outer skin in Step 05 fixes into each brace.
- 4 If a brace can move without bearing at its ends, recut it. Do not substitute the face X trim, a thin strap or screws into end grain for a fitted compression brace.

DONE WHEN

Both halves have snug braces, every member is in the same plane, and the door remains square. Confirm the brace / skin connection at CHECK B.

05 / Cut & screw on the outer skin

Part B + G. Four sheets for the pair at the reference width; horizontal joints simplify the frame.



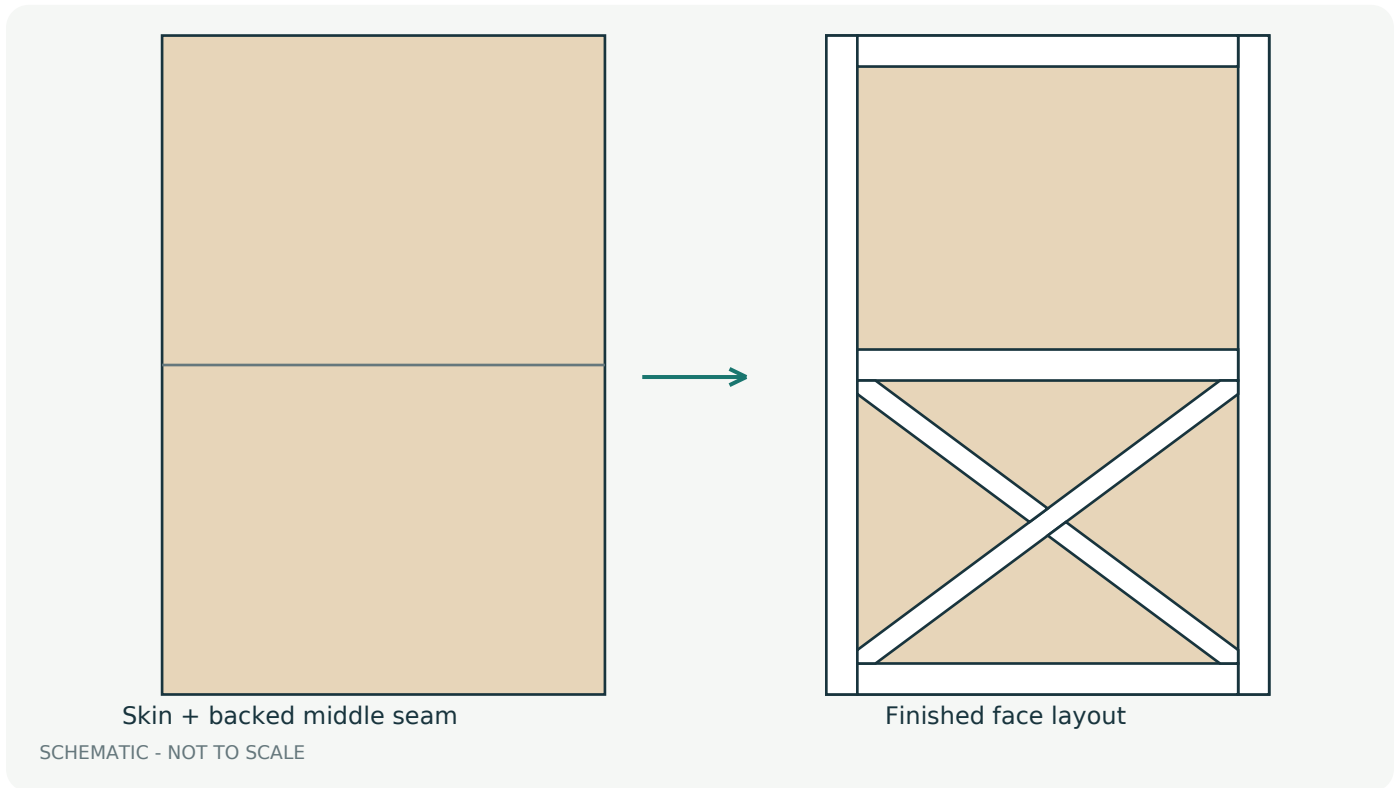
- 1 Confirm the panel stamp says Exterior. Cut four half-panels to the worksheet, with finished faces outward. Seal fresh panel edges with the compatible primer and let dry.
- 2 Position each pair with the confirmed joint gap centered on the middle rail. Keep the outside edges flush. Mark the stiles, all rails and braces on the skin so every fixing reaches wood.
- 3 Planning fastening pattern: #8 x 1-1/4 in. exterior screws at 150 mm along panel perimeter / middle joint and 200 mm along braces. Predrill, keep screw centers at least 10 mm from panel edges, and avoid splitting the frame. CHECK B must accept the final pattern.
- 4 Drive heads just flush, not through the face veneer. Fasten from the middle outward while the frame stays flat. Do not glue across the plywood expansion joint.

DONE WHEN

All panel edges are backed by the frame and every brace is connected to the skin. No unsupported seam or missed framing. Do not treat this draft pattern as a wind-load design.

06 / Fit the carriage-style trim

Part D + H + L. Trim is decorative; keep the internal bracing unchanged.



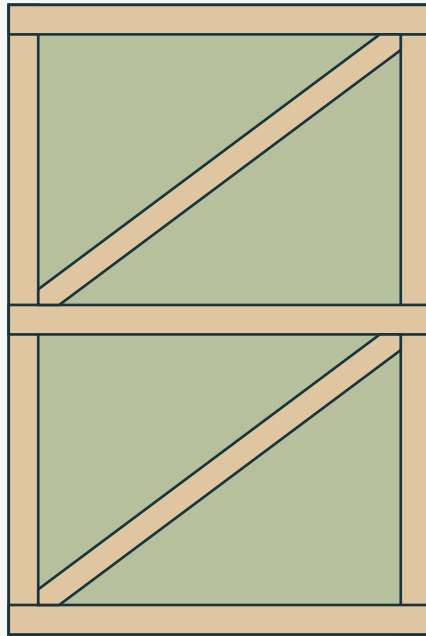
- 1 Prime trim backs and all ends before fitting. Install the two vertical edge trims, then the top, middle and bottom horizontal trims between them. Place the middle trim over the sheet joint.
- 2 Scribe the lower X to the real opening. Use one full diagonal and two infill pieces butted to it, all in one plane. Do not stack the two boards at the crossing.
- 3 Predrill and fix straight trim at about 300 mm centers using #8 x 2 in. screws: about 20 mm into the frame with 19 mm trim + 12 mm skin. For the X, mark fixings over braces; use the 2 reserved boards for frame-connected blocking behind other fixing points before insulating.
- 4 At the middle seam, install a compatible flexible flashing strip behind the cover rail, bridging the panel gap as its instructions permit. Seal the rail top / ends and preserve drainage below. Slope or cap horizontal trim to shed water; prime all fresh cuts.

DONE WHEN

The X is flat, trim screws reach solid wood, the middle seam is protected, and water can drain at the base. The finish / flashing detail is part of the pre-build review.

07 / Insulate & close the inside

Parts E + C + G. Use rigid stone wool, not an exposed foam substitute.



38 mm cavity

SCHEMATIC - NOT TO SCALE



Removable plywood cover

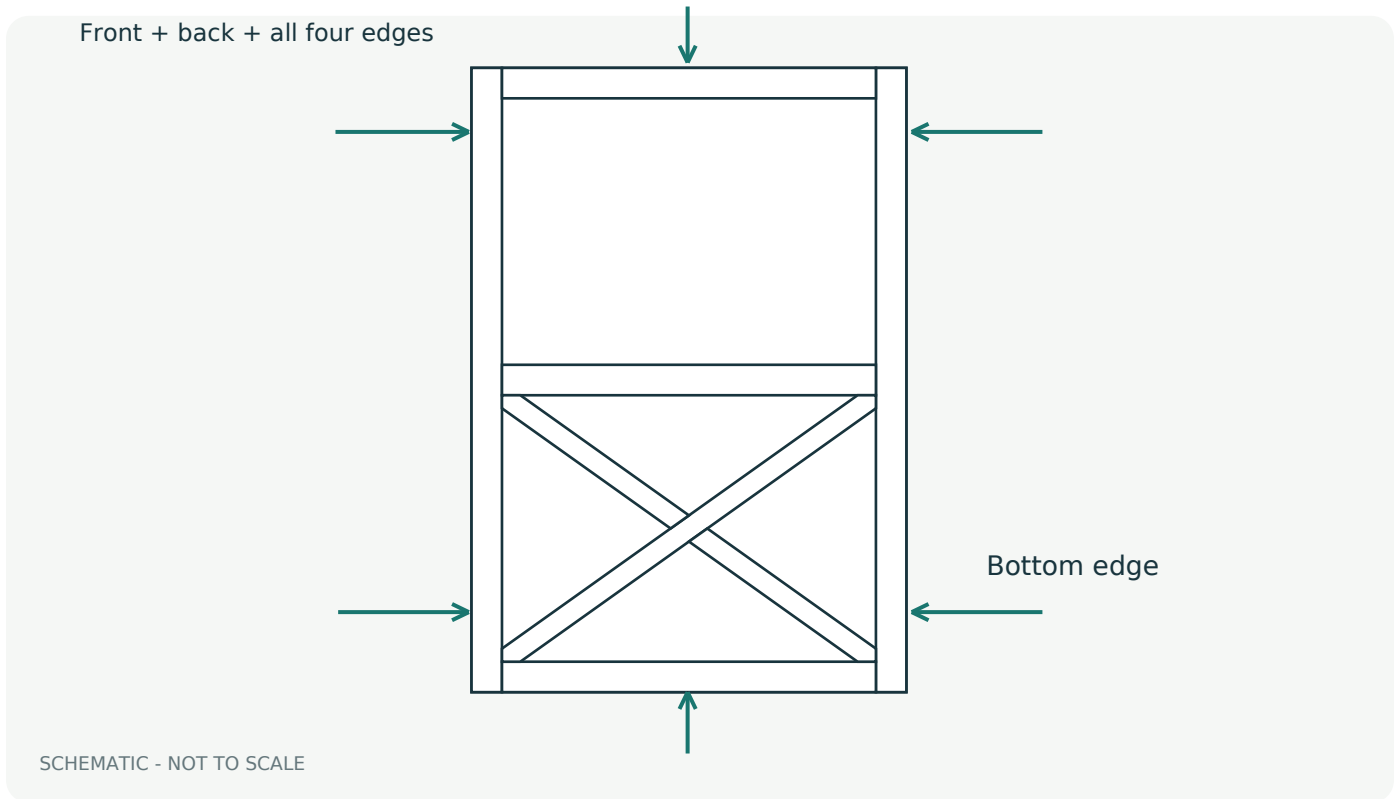
- 1 Finish / seal the accessible cavity wood and the back of the exterior plywood after the frame glue has cured. Let it dry fully before insulation; keep rain out of the work area.
- 2 Cut 1-1/2 in. (about 38 mm) Comfortboard to each actual triangular cavity. Fit snugly without bowing the lining; do not pack wet material or force excess thickness into the frame.
- 3 Make a cut layout before opening the pack. The 48 sq ft allowance is based on the smaller cavity area after subtracting the frame and braces; verify offcut reuse and waste.
- 4 Prime both sides / edges of the inner plywood, then screw it to the frame and braces at the Step 05 planning spacing. Keep the middle joint backed and hinge nuts accessible. Do not glue the lining; fully contain the mineral fibers.

MATERIAL LIMIT

Stone wool is noncombustible; the finished timber door is not fire-rated. Comfortboard is normally wall sheathing, not a tested door kit. Confirm this custom use and moisture detail; do not claim a whole-door R-value.

08 / Finish every exposed surface

Parts J + K + L. Protect the bottom edge especially carefully.



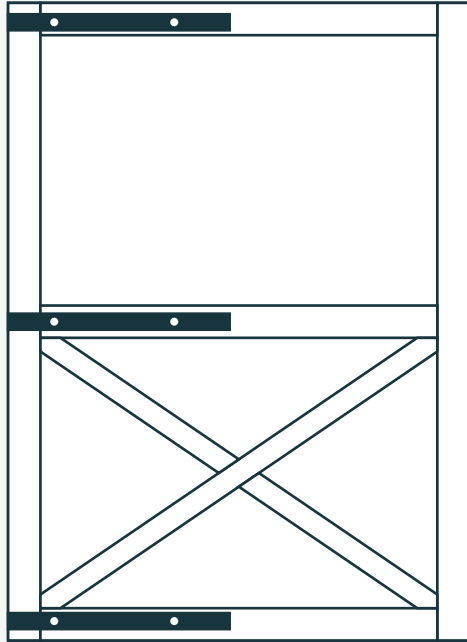
- 1 Lightly smooth new timber and veneer with suitable abrasives and dust extraction. Ease sharp edges lightly so paint can wrap them. Do not sand the old painted doors as part of this step.
- 2 Prime bare wood, cuts, filled holes and fresh hardware holes. Follow the primer label for ventilation, temperature, recoat timing and oily / solvent waste handling.
- 3 Apply two compatible exterior finish coats to front, back and all four edges, including behind removable hardware where practical. Keep pivot surfaces, gasket contact faces and drains free of paint buildup.
- 4 Let the coating cure sufficiently before stacking, hanging or compressing seals. Recoat any cut / drilled surface exposed during installation.

DONE WHEN

No raw end grain or panel edge remains exposed. Paint is a maintained protective finish, not a substitute for drainage or clearance above wet paving.

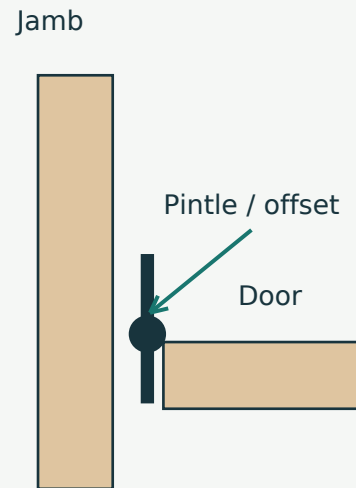
09 / Confirm & dry-fit the hardware

CHECK C must be complete. The 24 in. strap link is a candidate component, not a ready-to-install hinge set.



Three shown; final count / positions by supplier

SCHEMATIC - NOT TO SCALE



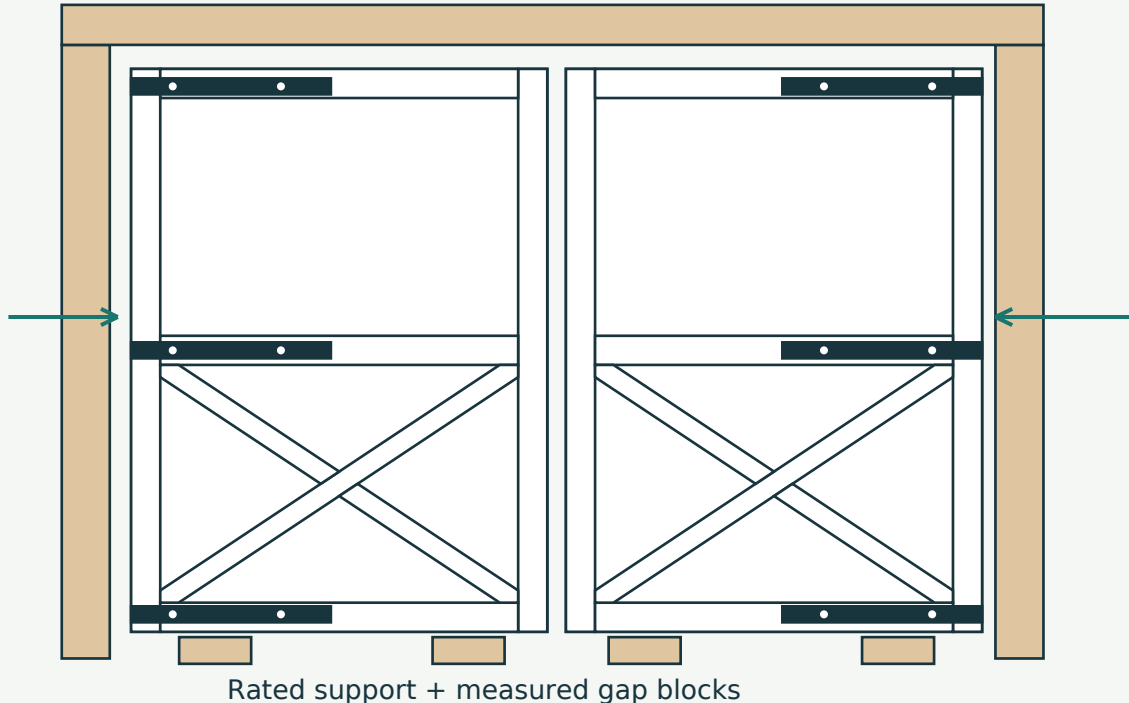
- 1 Verify the preselected hinge system against the finished leaf weight, including hardware. For early planning only, allow roughly 60-70 kg per leaf; actual products can differ. Never choose hinges from this estimate alone.
- 2 Confirm matched straps and pintles, corrosion protection, required hinge count / spacing, opening angle, anti-lift retention and all bolt / washer / backing requirements. Do not add individual catalog load figures together.
- 3 Dry-fit hardware to one leaf on the bench. Straps must bear on supported, coplanar mounting areas and connect to structural timber, not just the decorative skin. Add reviewed blocking where needed.
- 4 Transfer holes with the supplied template. Use the specified diameters, distances and corrosion-compatible fasteners; protect drilled wood. Leave installation to the approved jamb detail in Step 10.

STOP IF UNRESOLVED

The 120 / 140 mm timber face widths do not establish jamb depth or soundness. No generic lag length, pintle size, bolt pattern or wind capacity is authorized by this manual.

10 / Remove the old leaves & hang

Do this only after the new pair is ready and the hazards / jamb / hardware checks are closed.



SCHEMATIC - NOT TO SCALE

- 1 With helpers and rated lifting / support equipment, secure each old glazed leaf before releasing any hardware. Remove only the leaves and obsolete door hardware; protect the opening and retain all structural timber.
- 2 Inspect the newly exposed mounting timber. Soft wood, cracking, loose joints or movement stops installation. Do not cover damage with a new mounting block or longer screws.
- 3 Place the passive left leaf on firm supports at its released height. Plumb the hinge axis, establish head / side gaps with blocks, then attach the matched hardware using the approved detail.
- 4 Support and hang the right leaf to match. Keep supports in place until all connections are secure. Open slowly through the full planned swing, then check gaps without lifting the latch edge.

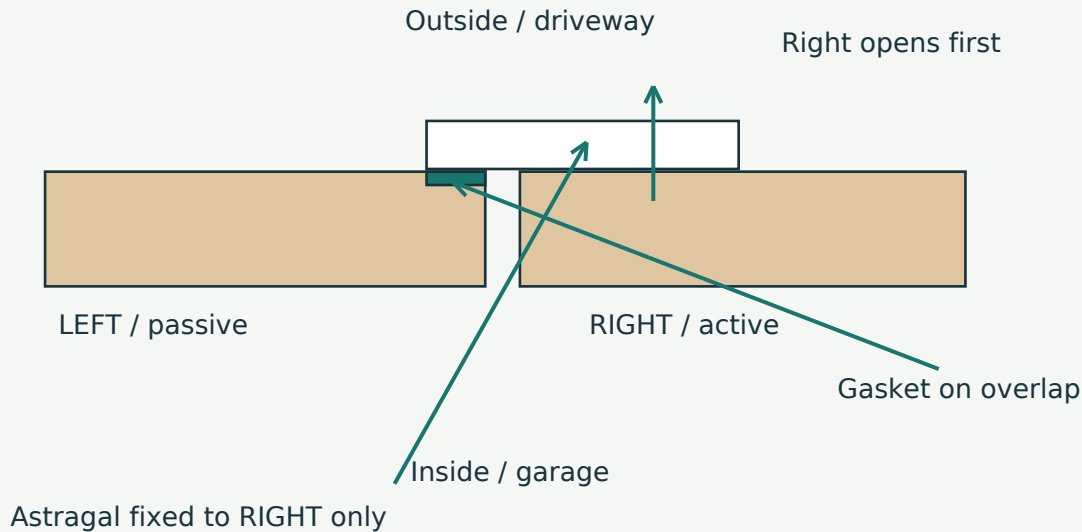
DONE WHEN

Both leaves swing without rubbing or lifting, the hinge axes are stable, and supports can be removed safely. Work in calm conditions; never leave an unsecured leaf free to swing.

11 / Add the stops, astragal & seals

Parts D + P + Q + R. Close the left leaf first; close the right leaf second.

TOP VIEW / MEETING EDGE



SCHEMATIC - NOT TO SCALE

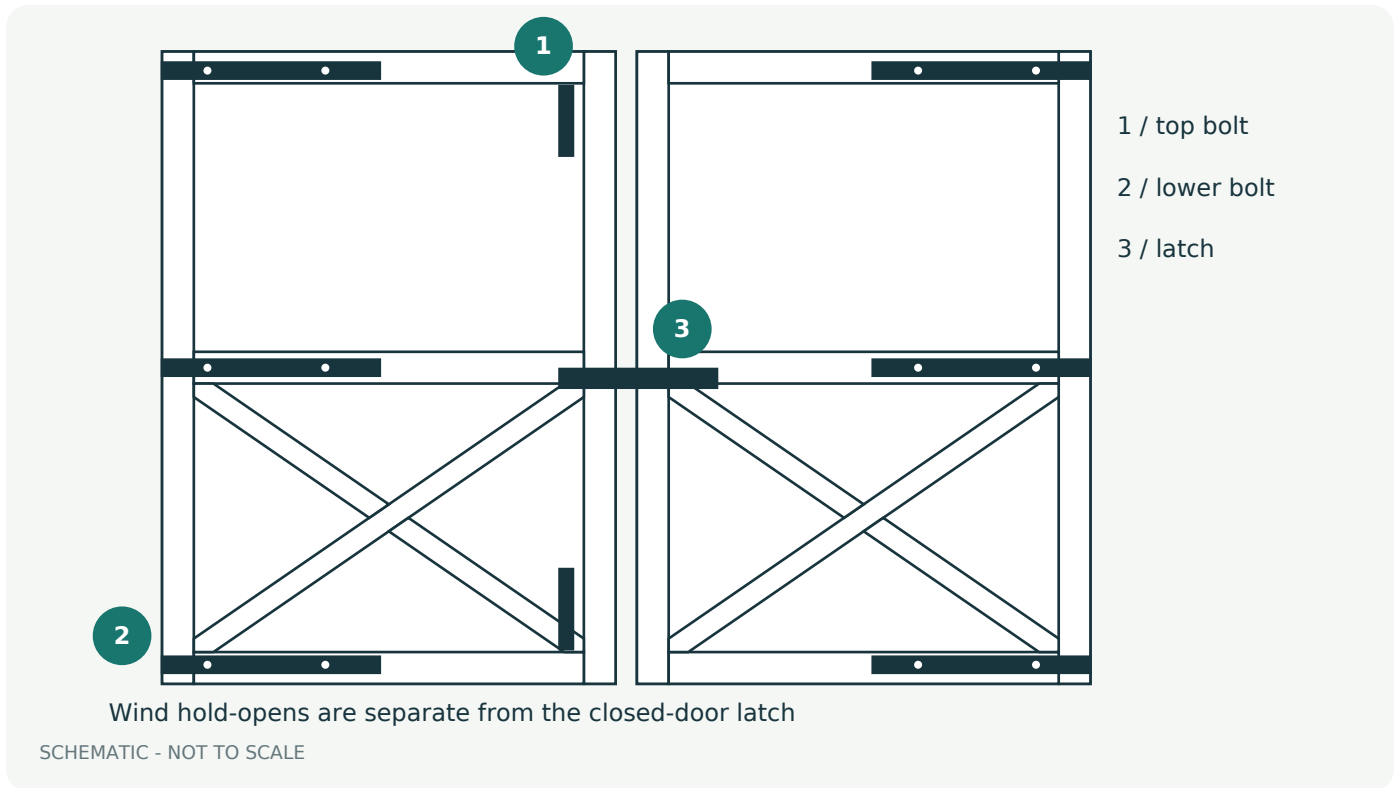
- 1 Fit inside jamb / header stops so closed doors contact a compressible seal. An 8 ft board is shorter than this header: use two stop sections, each fixed to continuous sound backing, with a sealed joint. Stops must clear the outward swing and straps.
- 2 Fit a full-height exterior astragal to the right active leaf only. Field-size its overlap and gasket so it covers the meeting gap and compresses against the left leaf without binding.
- 3 Fit continuous bottom sweeps long enough for each leaf. Confirm a smooth, drainable threshold contact surface and that the sweep clears the entire swing. Do not join short sweeps across the opening or make asphalt carry a structural anchor.
- 4 Check weatherstrip contact at corners and intersections. Adjust stops and seals rather than forcing a stiff latch. Keep the outside drainage path open; do not create a water dam at the threshold.

DONE WHEN

A light paper strip feels consistent resistance around the closed perimeter. Right leaf opens first and closes last. Seal profile / compression and bottom gap are actual product-dependent dimensions.

12 / Fit the latch & wind restraints

Parts S + T + U + V. All hardware connects to solid, reviewed mounting points.



- 1 Secure the passive leaf with the reviewed top and bottom bolt arrangement. The linked cane bolt is only a candidate: confirm it is suitable for each orientation and remains captured / restrained when retracted.
- 2 Use appropriate receivers in verified sound structure. Before any slab drilling, establish concrete integrity, edge distances and hidden services. If these are unknown, stop that fixing and obtain a suitable alternative detail.
- 3 Fit the active-leaf latch and pulls to backed timber. Confirm the astragal does not obstruct the mechanism and that a person inside can release it without being trapped. Coordinate with any required exit route.
- 4 Install a positive hold-open for each leaf at its planned open position, using supplier-approved anchorage. A friction kick-down stop or loose weight is not a wind restraint; close and secure the doors before strong winds.

DONE WHEN

Bolts fully engage, the latch works without lifting the leaf, inside release works, and neither open leaf can swing freely. These details do not create a tested wind rating.

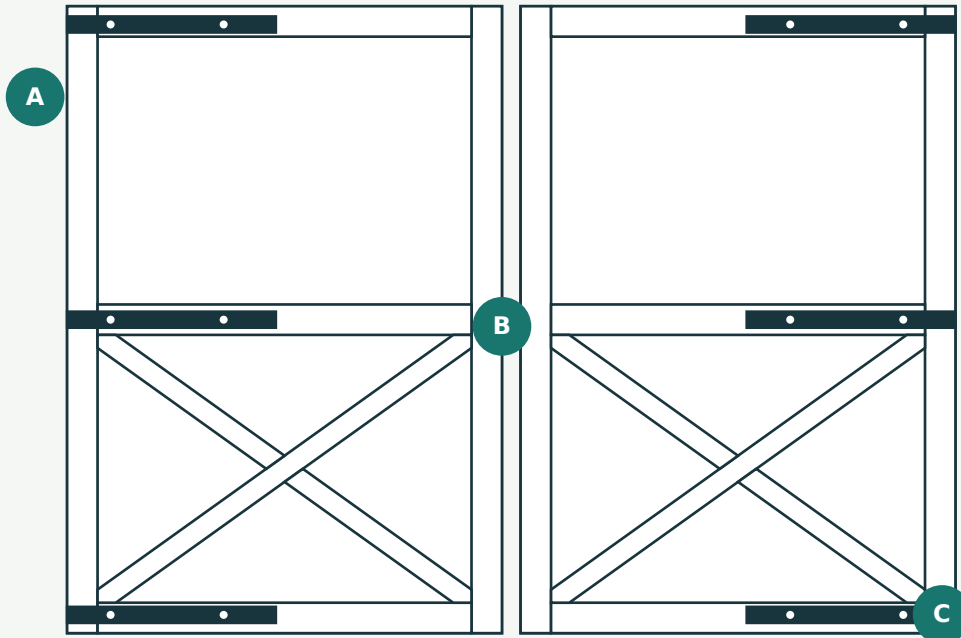
13 / Test, adjust & maintain

Complete the check before normal use; keep this page with your final hardware instructions.

A / stable hinges

B / easy closing

C / dry base



SCHEMATIC - NOT TO SCALE

- 1 Cycle both leaves slowly at least ten times. Check rubbing, shifting hardware, loosening joints, latch alignment and the right-first opening sequence. Stop if movement or splitting develops.
- 2 Inspect the sealed perimeter from inside with daylight outside. Check again after normal rain; identify the entry path before adding sealant. Do not pressure-wash the door to test it.
- 3 Record finished leaf weights, final gaps, hinge model, fixings and photographs. Complete any required inspection. Keep all product instructions with the manual.
- 4 Recheck after the first week and after the first wet / cold weather change. Inspect seasonally: bottom paint edges, panel seams, hinges, bolts, hold-opens and moisture behind the removable lining. Repair worn finish before raw wood becomes exposed.

HEATING LIMIT

This insulated door alone does not make the garage an insulated room. Use heating equipment only as its instructions permit; never use grills, generators or vehicle exhaust for heat inside the garage.

References & purchase notes

Sources checked 12 Sep 2026. Product names on pages 1-2 are clickable Home Depot purchase links.

1 / Project scope

Two new left-bay leaves, no reused door parts. Current pair 257 x 216 cm is user-supplied. Center post 14 cm; left / top timber face widths 12 cm. Depth, condition, clear opening and diagonals remain unverified.

2 / Home Depot product listings

Prices marked P were publicly displayed product / category prices. No selected Westfield store, local stock, delivery or checkout price was verified. Plywood prices matched models 201429 and 235552. Timber / insulation / long sweeps / hold-opens include explicit allowances.

3 / ROCKWOOL Comfortboard

Manufacturer describes a noncombustible stone-wool sheathing board. Its use between moving custom-door skins is a design proposal, not a manufacturer-certified door system. Nominal board insulation performance is not whole-door performance.

4 / APA Sanded Plywood Guide

Confirm the actual panel classification and installation information. Exterior and Exposure 1 are different bond classifications; this weather-exposed outer skin specifies Exterior. Joint spacing and fastening must suit this application.

5 / Titebond III technical data

The manufacturer excludes structural / load-bearing use and directs exterior-door users to technical support. Confirm or replace this adhesive before assembly. Current instructions govern temperature, working time, clamping and cure.

6 / National Hardware strap candidate

N248-047 is a strap component. Obtain written confirmation of its matching hook / pintle, door-width and weight basis, hinge count, offsets and attachment requirements. Catalog safe-working-load figures are not a completed-door approval.

Release sheet / questions together

Fill this once. No need for another general photo survey; only targeted checks to close the gaps below.

Field	Record
Clear widths: top / middle / bottom	_____ / _____ / _____ mm
Clear heights: left / center / right	_____ / _____ / _____ mm
Opening diagonals: D1 / D2	_____ / _____ mm
Jamb depth / condition; swing high point	_____
Released leaf L x D; gL/gR/gM/gT/gB	_____
Final hinge + pintle + bolt / anchor detail	_____
Finished weight: left / right	_____ / _____ kg
Frame / fastening review, name + date	_____
Local requirements / inspection decision	_____

Questions to resolve before release

What are the final corrected opening and swing-clearance dimensions? Which complete hinge / attachment system is accepted for the finished leaf weight and jambs? Does local review require any change to this custom assembly or its installation?

[Westfield Building & Construction](#) - confirm requirements for replacement leaves; do not assume the separate right-bay alteration has the same permit scope.

[EPA lead-safe DIY guidance](#) - assess old paint before disturbance; use the appropriate lead-safe process. [EPA asbestos guidance](#) - leave suspect siding / sheet materials undisturbed pending assessment.

READY FOR THE NEXT REVISION

Once the measurements and review details are available, replace the blanks with a released cut list and final hardware schedule. The numbered assembly sequence can remain the same unless the reviewed design changes.